

Curriculum Vitae

Bolong Liu, Ph.D.

1. BASIC INFORMATION

Name	Bolong Liu	Profile Photo	
Current Position	Associate-Senior Professional Title, NUDT		
Affiliation	College of Intelligence Science and Technology, NUDT; School of Mechanical Engineering, BIT		
E-mail	liubolong_15@163.com	ORCID	https://orcid.org/0000-0003-2683-0515
Scopus Author ID	57224313257	Application Journal	Ocean Engineering

2. RESEARCH PROFILE AND JOURNAL FIT

Dr. Bolong Liu works on digital design, hydrodynamic optimization, energy-saving mechanisms, and intelligent control of cross-domain platforms. His research focuses on vehicles operating across water, land, and air, with a systematic research foundation in water-land amphibious platforms and a growing research direction in water-air cross-domain platforms.

His work integrates hydrodynamics, CFD-based analysis, model-test validation, data-driven configuration optimization, and intelligent control. This research profile is closely aligned with Ocean Engineering in special marine vehicle design, resistance and propulsion, energy-efficient devices, high-fidelity simulations, model-test validation, and intelligent ocean engineering systems.

3. RESEARCH INTERESTS

- Cross-domain platforms and special marine vehicles
- Water-land amphibious platforms and water-air cross-domain platforms
- Hydrodynamic optimization, drag reduction, resistance and propulsion
- Energy-saving technologies for marine and cross-domain vehicles
- CFD-based design, model-test validation, and fluid-structure interaction
- Data-driven configuration optimization and intelligent control

4. EDUCATION AND WORK EXPERIENCE

1. B.Eng. in Mechanical Engineering and Automation, National University of Defense Technology, 2015-2019.
2. Ph.D. in Mechanical Engineering, National University of Defense Technology, 2019-2024. Supervisor: Prof. Xiaojun Xu.
3. College of Intelligence Science and Technology, National University of Defense Technology, 2024-Present. Associate-senior professional title.
4. School of Mechanical Engineering, Beijing Institute of Technology, 2025-Present. Postdoctoral research under the supervision of Prof. Changle Xiang, Member of the Chinese Academy of Engineering.

5. ACADEMIC OUTPUTS AND PROFESSIONAL SERVICE

- Published more than 30 academic papers, including over 20 SCI-indexed papers, with representative work in Ocean Engineering, Journal of Cleaner Production, Physics of Fluids, and Ships and Offshore Structures.
- Authorized 8 Chinese invention patents related to mobile platforms, amphibious vehicles, configuration design, and control methods.
- Reviewer for Ocean Engineering, Physics of Fluids, Ships and Offshore Structures, and other SCI-indexed journals in marine engineering, hydrodynamics, and intelligent vehicle systems.
- Currently leading collaborative research efforts between National University of Defense Technology and Beijing Institute of Technology, with several manuscripts under preparation for topics closely aligned with Ocean Engineering.

6. SELECTED RESEARCH PROJECTS

1. Principal Investigator, National Natural Science Foundation of China project on data-driven high-speed configuration optimization of amphibious platforms.
2. Principal Investigator, Basic Strengthening Program project related to cross-domain platform design and optimization.
3. Principal Investigator, institutional autonomous research fund project on wheel-legged amphibious unmanned platforms in complex water environments.
4. Participant or core contributor to multiple projects supported by the National Natural Science Foundation of China, the Ministry of Industry and Information Technology of China, and other national or ministerial-level programs.
5. Sub-project contributor to a MIIT key project on multifunctional emergency rescue platforms for complex water-land scenarios.

7. POTENTIAL CONTRIBUTIONS TO OCEAN ENGINEERING

1. Provide timely, rigorous, and constructive reviews in hydrodynamics, CFD, special marine vehicles, cross-domain platforms, energy-saving technologies, and data-driven design optimization.
2. Recommend qualified reviewers from the marine engineering, intelligent vehicle, and cross-domain platform communities.
3. Help attract high-quality submissions from China and the broader Asia-Pacific research community.
4. Promote Ocean Engineering among researchers working on energy-efficient marine vehicles, CFD-based optimization, and intelligent ocean engineering systems.
5. Support emerging topics such as data-driven hydrodynamic design, intelligent cross-domain platforms, and water-air cross-domain vehicle systems.

8. SELECTED PUBLICATIONS

1. Liu B, Wang J, Pan D, et al. Energy-saving performance of formation sailing strategies for amphibious vehicles: a hydrodynamic and energy perspective. *Journal of Cleaner Production*, 2026, 546: 147771.
2. Liu B, Pan D, Xie Y, et al. Research on drag reduction and energy saving of M-bow amphibious vehicle based on configuration optimization. *Ocean Engineering*, 2025, 342: 123161.
3. Liu B, Pan D, Xu X. Research on the resistance and maneuvering characteristics of an amphibious transport vehicle and the influence of stern hydrofoil. *Ocean Engineering*, 2024, 293: 116592.
4. Liu B, Xu X, Pan D. Resistance reduction optimization of an amphibious transport vehicle. *Ocean Engineering*, 2023, 280: 114854.
5. Liu B, Xu X, Pan D. Influence of resistance due to locomotion mechanism configurations of a new high-speed amphibious vehicle (HSAV-II). *Ocean Engineering*, 2023, 283: 115175.
6. Liu B, Zhang Y, Pan D, et al. Amphibious vehicle's resistance optimization through neural networks and genetic algorithms. *Physics of Fluids*, 2024, 36(6): 065129.
7. Liu B, Pan D, Xu X. Study and optimization of a deformable trimaran amphibious transport vehicle's resistance characteristic. *Physics of Fluids*, 2024, 36(4): 045159.
8. Liu B, Xu X, Pan D. Research on launching, water exiting, and river crossing of an amphibious vehicle. *Physics of Fluids*, 2023, 35(11): 113328.
9. Liu B, Xu X, Pan D, et al. Research on shipping energy-saving technology: Hydrofoil amphibious vehicle driven by waterjet propulsion. *Journal of Cleaner Production*, 2023, 382: 135257.
10. Liu B, Xu X, Pan D, et al. Drag reduction design and research of high-speed amphibious vehicle's deformable track wheels. *Ships and Offshore Structures*, 2023, 18(7): 970-979.

11. Pan D, Xu X, Liu B, et al. A review on drag reduction technology: Focusing on amphibious vehicles. *Ocean Engineering*, 2023, 280: 114618.
12. Pan D, Xu X, Liu B. Numerical analysis on hydrodynamic performance and hydrofoil optimization for amphibious vehicles. *Physics of Fluids*, 2023, 35(8): 083330.
13. Pan D, Xie Y, Liu B, et al. Experimental investigation on hydrodynamic performance of amphibious vehicles with hydrofoil in calm water and regular waves. *Ships and Offshore Structures*, 2025: 1-17.
14. Xu X, Liu B, Pan D, et al. Influence of towing position on hydrodynamic characteristics of an Amphibious Tractor. *Ocean Engineering*, 2023, 272: 113932.